

Machine Learning for teens

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Ways to Overcome Overfitting

- ▶ **Feature Selection**
 - ▶ Simplify the model by removing irrelevant or noisy features.
- ▶ **Regularization**
 - ▶ Regularization works for linear models.
- ▶ **Cross-Validation**
 - ▶ KFold, StratifiedKFold, cross_val_score
- ▶ **Model Constraints** (for Trees)
 - ▶ Like our code
- ▶ **Dimensionality Reduction**
 - ▶ Reduce feature space like PCA

Ways to Overcome Overfitting

- ▶ **Dropout**
 - ▶ For neural networks
- ▶ **Data Augmentation (for images/text)**
 - ▶ Increase training diversity.
 - ▶ Flip, rotate, crop images
 - ▶ Synonym replacement for text
- ▶ **Ensemble Methods**
 - ▶ Combine models to reduce variance.
 - ▶ Bagging ,Boosting and Stacking

Difference between bias and variance

- ▶ Bias
 - ▶ The model doesn't follow the original data.
 - ▶ It makes big mistakes because it's too simple.
 - ▶ It ignores the real curve and gives wrong predictions.
 - ▶ Result: **Underfitting**
- ▶ Variance
 - ▶ The model follows the original data too closely, even the noise.
 - ▶ It learns every tiny detail – even things that don't matter.
 - ▶ So it works only on training data, but fails on new data.
 - ▶ Result: **Overfitting**
- ▶ Bias = too simple → misses the pattern
- ▶ Variance = too detailed → memorizes noise

Difference between bias and variance

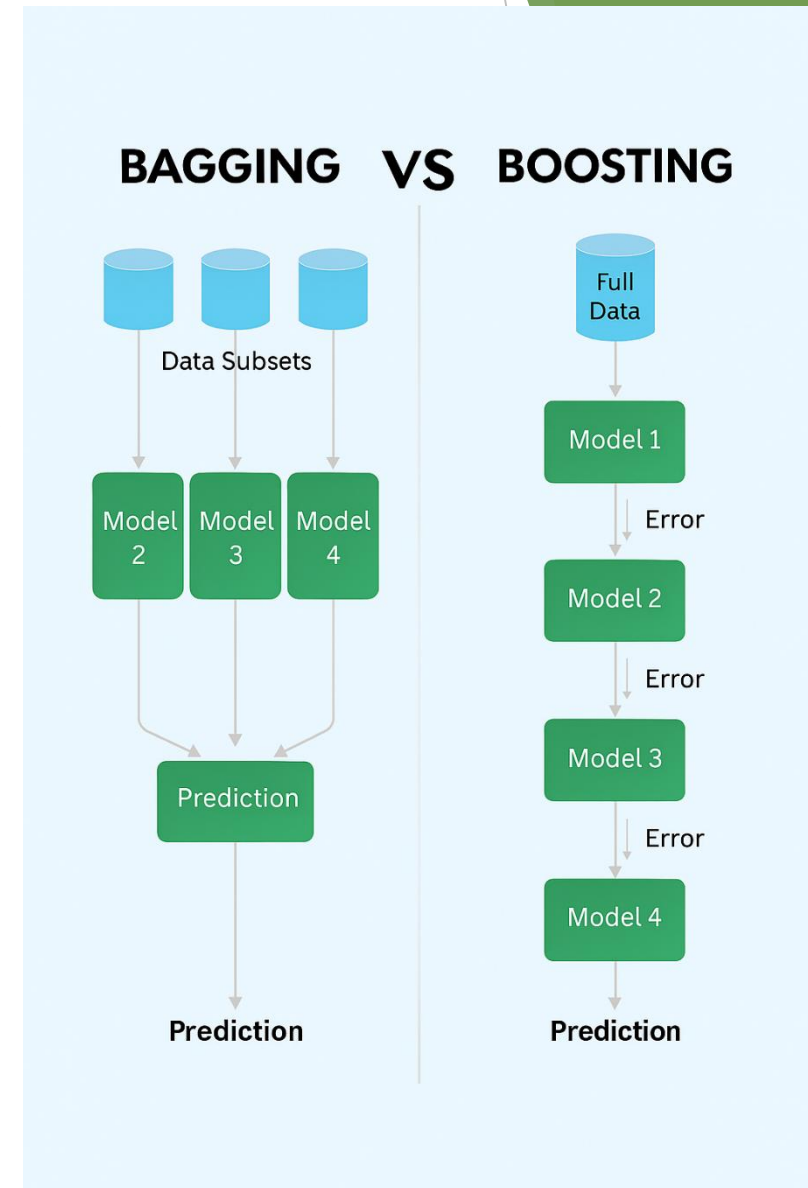
- ▶ The **bias-variance trade-off**
 - ▶ decrease bias, variance tends to increase
- ▶ In practice
 - ▶ we want a model that has low bias and low variance

Ensemble methods

- ▶ Ensemble methods are like **teamwork for machine learning models**.
- ▶ we use a **group of models** and combine their answers to get a better result

Bagging vs Boosting

- ▶ **Bagging**
 - ▶ multiple models trained in parallel on different data subsets.
- ▶ **Boosting**
 - ▶ models trained sequentially,
 - ▶ each correcting the errors of the previous one.



Boosting for overcoming underfitting

- ▶ Boosting
 - ▶ can help reduce **bias**
 - ▶ good at overcoming underfitting
 - ▶ can go too far and cause **overfitting**
 - ▶ Regularization
 - ▶ early stopping (model constraint)
 - ▶ small learning rates

Types of boosting

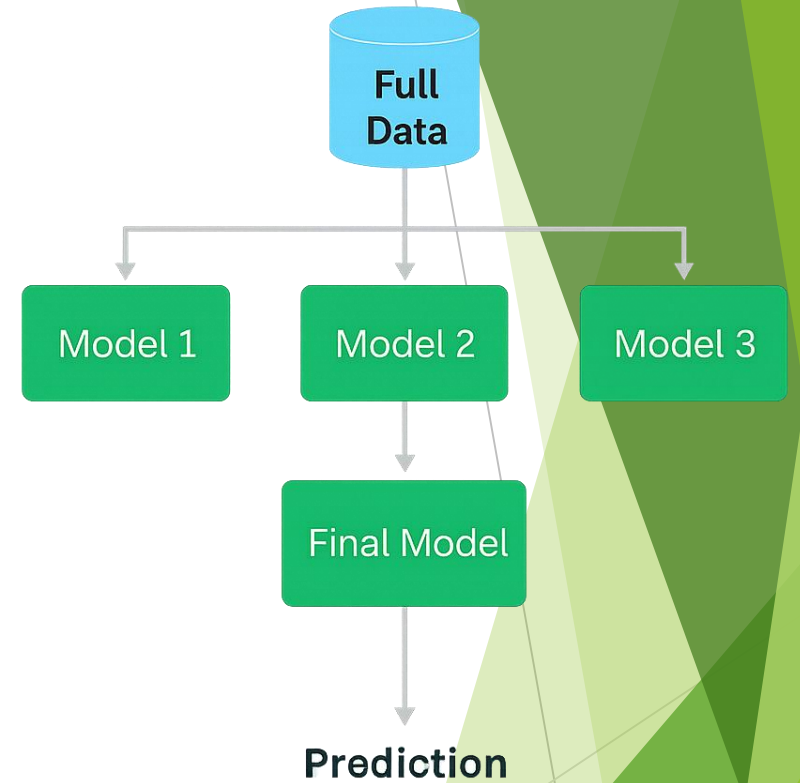
Algorithm	Strengths	Best Use Case
AdaBoost	Simple, interpretable	Clean, small datasets
GradientBoost	Flexible, powerful	General-purpose boosting
XGBoost	Fast, regularized, scalable	Large datasets, competitions
LightGBM	Very fast, efficient	Big data, high performance
CatBoost	Handles categorical features well	Tabular data with categories

Stacking

- ▶ Stacking (or stacked generalization) is a two-layer ensemble technique
 - ▶ Level-0 (Base Models)
 - ▶ Multiple models (e.g., decision tree, logistic regression, SVM) are trained on the same dataset. Each learns in its own way.
 - ▶ Level-1 (Meta-Model)
 - ▶ A final model (often a simple one like logistic regression) takes the predictions of the base models as input and learns how to best combine them.

Stacking vs Boosting

- ▶ Stacking and boosting are both ensemble learning methods
 - ▶ Boosting
 - ▶ builds models sequentially
 - ▶ stacking
 - ▶ trains diverse models in parallel and uses a meta-model to learn



Comparison of Ensemble Learning Methods

Feature	Bagging	Boosting	Stacking
Training Style	Parallel	Sequential	Parallel
Base Models	Same type	Same type	Different types
Data Used	Bootstrapped subsets	Full data	Full data

Assignment

- ▶ Compare Ensemble Methods
 - ▶ Use student-data.csv
 - ▶ Apply all ensample methods